



Name _____ Period _____ Test Day _____

Unit 11 - Gas Laws Essential Knowledge and Guided Notes

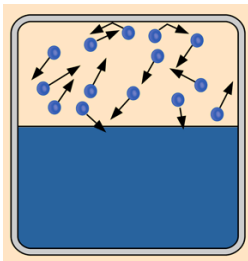
Textbook: CK Chemistry Chapter 13 - States of Matter and Chapter 14 - Gas Laws
<https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Helpful Videos and Practice: Posted on CANVAS

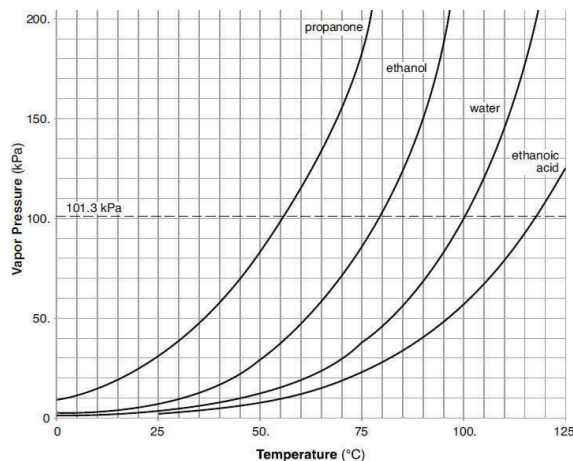
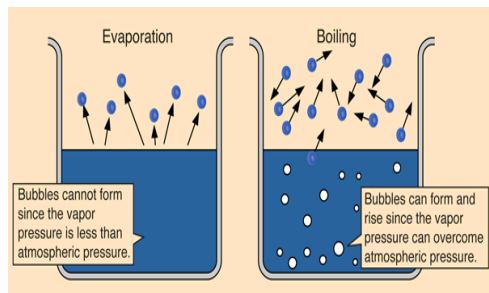
Essential Knowledge:

1. Gases have mass and occupy space. Gas particles are in constant, rapid, random motion and exert pressure as they collide with the walls of their containers. Gas molecules with the lightest mass travel fastest. Relatively large distances separate gas particles from each other.
2. Pressure units include atm, kPa, and mmHg. Volume units include mL, L, cm^3 , dm^3 . Temperature units include Kelvin (K) and Celsius ($^{\circ}\text{C}$). Absolute zero on the Kelvin scale is where all kinetic energy ceases. $\text{K} = 273 + ^{\circ}\text{C}$
3. Vapor pressure is the pressure of the vapor found directly above a liquid in a closed container. When the vapor pressure equals the atmospheric pressure, a liquid boils.
4. Volatile liquids have high vapor pressures, weak intermolecular forces, and low boiling points. Nonvolatile liquids have low vapor pressures, strong intermolecular forces, and high boiling points.

Vapor Pressure:

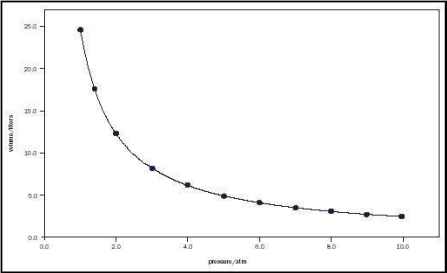
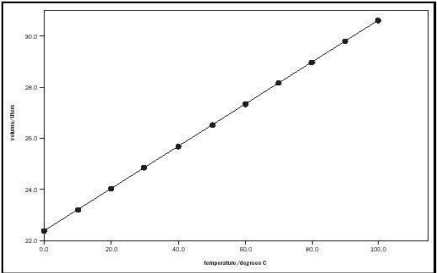
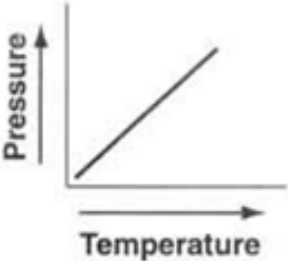


Evaporation vs. Boiling



The effect of temperature on vapor pressure

5. Gas Laws allow you to predict changes in pressure, volume, number of moles and temperature of gases under "ideal" conditions. *Note: Ideal gases don't exist, but these laws work well in making predictions under conditions of low pressure and high temperature when intermolecular forces are least influential. You will need to learn the ones on the next page.*

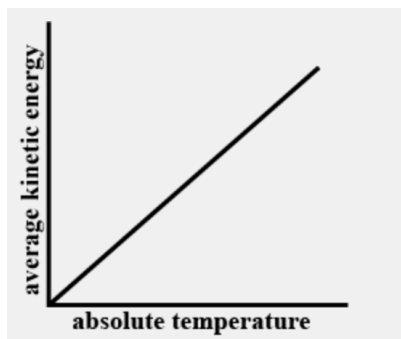
Gas Laws	Notes
Boyle's Law (Pressure vs. Volume) Inverse $P_1V_1 = P_2V_2$ At constant temperature, the pressure and volume of a sample of a gas are inversely proportional to each other.	
Charles' Law (Temperature vs. Volume) Direct $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ At constant pressure, the volume of a fixed amount of gas is directly proportional to its absolute temperature.	
Gay-Lussac's Law (Temperature vs. Pressure) Direct $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ At constant volume, the pressure of a fixed amount of gas is directly proportional to its absolute temperature	
Combined Gas Law	$(P_1V_1/T_1 = P_2V_2/T_2)$ relates pressure, volume, and temperature of a gas.
Dalton's Law of Partial Pressure The sum of the partial pressures of all the components in a gas mixture is equal to the total pressure of a gas mixture	$P_T = P_a + P_b + P_c$ where a, b, c are different gases in the same container. $P_a = (X_a)(P_T)$ Where X_a is the mole fraction of a = (moles_a)/(moles_T)
Ideal Gas Law	$PV = nRT$ P = pressure, V = volume, n = number of moles, T = Temp in K and R is a constant that is dependent on units. The most common R values are: 0.0821 (L atm)/ mol or 8.314 (L kPa)/ mol K)

U11 Lesson #1 Units, Vapor Pressure, and Boiling Point

Variables that affect gases: _____, _____, _____, _____

Temperature:

Definition: a measure of the kinetic energy of particles (motion) As temperature increases so does the kinetic energy. It is a direct relationship.



Temperature Scales used in science: _____, _____ Formula for Converting: $K = ^\circ C + \underline{\hspace{1cm}}$

Try these Examples:

$$306 \text{ K} = \underline{\hspace{1cm}} ^\circ C$$

$$\underline{\hspace{1cm}} \text{ K} = 100 ^\circ C$$

$$223 \text{ K} = \underline{\hspace{1cm}} ^\circ C$$

Volume:

Units of Volume used in science: _____ liter = _____ mL = _____ cm^3 = _____ dm^3

Examples:

$$2.5 \text{ L} = \underline{\hspace{1cm}} \text{ mL} = \underline{\hspace{1cm}} \text{ cm}^3 = \underline{\hspace{1cm}} \text{ dm}^3$$

$$\underline{\hspace{1cm}} \text{ L} = \underline{\hspace{1cm}} \text{ mL} = \underline{\hspace{1cm}} \text{ cm}^3 = 0.50 \text{ dm}^3$$

$$\underline{\hspace{1cm}} \text{ L} = 750 \text{ mL} = \underline{\hspace{1cm}} \text{ cm}^3 = \underline{\hspace{1cm}} \text{ dm}^3$$

$$\underline{\hspace{1cm}} \text{ L} = \underline{\hspace{1cm}} \text{ mL} = 10400 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ dm}^3$$

Pressure:

What causes gas pressure? _____

What measures air pressure? _____

Units of Pressure used in science:

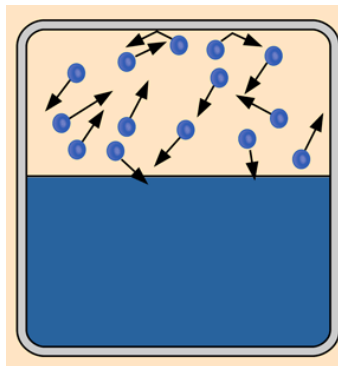
$$\underline{\hspace{1cm}} \text{ atm} = \underline{\hspace{1cm}} \text{ mmHg} = \underline{\hspace{1cm}} \text{ torr} = \underline{\hspace{1cm}} \text{ kPa}$$

Examples: (Hint: Use dimensional analysis to convert between units.)

$$0.250 \text{ atm} = \underline{\hspace{1cm}} \text{ mmHg} = \underline{\hspace{1cm}} \text{ torr} = \underline{\hspace{1cm}} \text{ kPa}$$

$$\underline{\hspace{1cm}} \text{ atm} = \underline{\hspace{1cm}} \text{ mmHg} = 579 \text{ torr} = \underline{\hspace{1cm}} \text{ kPa}$$

Vapor Pressure and Boiling Point: Read CAREFULLY!



Evaporation is a type of vaporization that occurs on the surface of a liquid as it changes into the gas phase.

If you **close** the system the vapor will begin to condense.

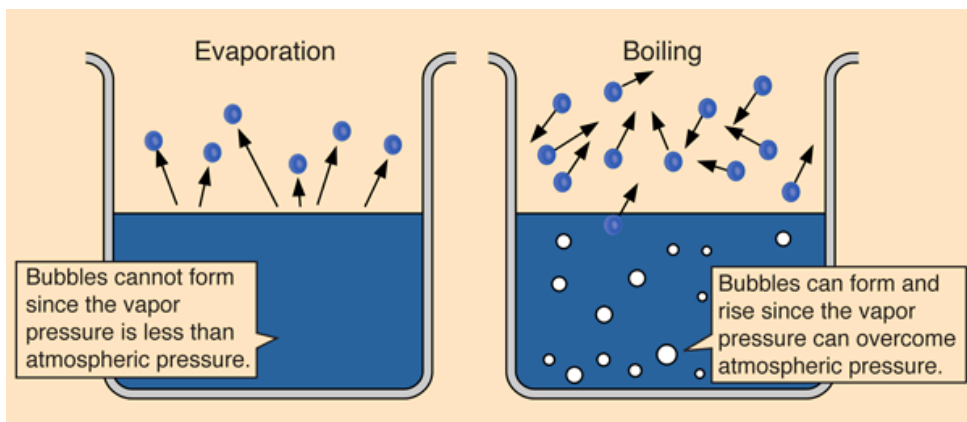
Vapor pressure occurs as the evaporated particles hit the sides of the container above the liquid.

The **Equilibrium Vapor Pressure** is exerted by a vapor in equilibrium (condensing and evaporation at the same rate) with its condensed phases at a given temperature **in a closed system**.

Liquids with _____ Vapor Pressures evaporate easily and have weak intermolecular forces.

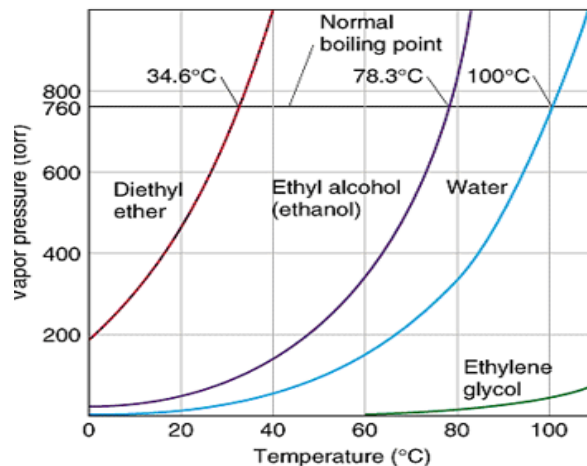
Liquids with _____ Vapor Pressures evaporate less easily and have stronger intermolecular forces

Definition of Boiling Point: Equilibrium Vapor Pressure = Outside Atmospheric Pressure
Bubbles can form and rise since the vapor pressure can overcome the atmospheric pressure.



Reading Vapor Pressure Graphs for Boiling Points

1. What is the pressure at the normal boiling point? _____. That is the same as _____ atm.
2. At what temperature does ethanol boil when the pressure is 400 torr? _____
3. At what temperature does Diethyl ether boil when the pressure is 400 torr? _____
4. Which has stronger intermolecular forces: Diethyl ether or ethyl alcohol? _____
5. Higher boiling points mean (stronger or weaker) intermolecular forces.
6. What temperature does water boil at 200 torr (200 mmHg)? _____



U11 Lesson #2 Basic Gas Laws

Four Variables you can measure about a gas:

_____, _____, _____, and _____

Boyle's Law

2 variables that change:

2 variables that stay constant:

In Words:

Graph Sketch:

This is a(n) _____ relationship

Equation:

Example:

If the initial pressure of a gas is 2.4 atm and the initial volume is 2.0 L what will the new volume be if I decrease the pressure to 0.80 atm?

Charles' Law

2 variables that change:

2 variables that stay constant:

In Words:

Graph Sketch:

This is a(n) _____ relationship

Equation:

Example:

If the initial temperature of a gas is 400. K and the initial volume is 2.0 L. What will the new volume be if I decrease the temperature to 100. K?

Gay Lussac's Law

2 variables that change:

2 variables that stay constant:

In Words:

Graph Sketch:

This is a(n) _____ relationship

Equation:

Example:

If the initial temperature of a gas is 200.K and has a pressure of 760. mm Hg and it is heated to 300.K, what is the new pressure assuming amount and volume are held constant?

Avogadro's Law 2 variables that change: 2 variables that stay constant: <u>In Words:</u>	Graph Sketch: This is a(n) _____ relationship	Equation: <u>Example:</u> If a container has 4.0 moles of gas at 1.0 atm, what will its new pressure be if 2.0 more moles of gas are added assuming constant temperature and volume?
Combined Gas Law Equation:	<u>Example:</u> - If a 3.00 L balloon ascends from standard temperature and pressure to higher altitudes, its new pressure is 720 mm Hg and the new temperature is -5 C, what is its new volume? $P_1 = 760 \text{ mm Hg}$ $V_1 = 3.00$ $T_1 = 273 \text{ K}$ $P_2 = 720 \text{ mm Hg}$ $V_2 = ??$ $T_2 = 268 \text{ K}$	

U11 Lesson #3 Ideal Gas Law and Dalton's Law of Partial Pressures

Real vs Ideal Gases

An _____ gas is assumed to have particles who exhibit no attractive or repulsive forces and whose particle size is insignificant. A _____ gas has intermolecular forces, particle volume and can change states.

So when do gas laws work? At _____ **temperatures** and _____ **pressures** where intermolecular forces are least effective.

Ideal Gas Law

Useful when 3 of the 4 variables are known. You can predict the other one.

$$PV = nRT$$

Where R is a constant that depends on the units chosen. It will always be given to you.

Most common units of R

R = _____ atm L/ mol K

or R = _____ kPa L/mol K

Example #1 - If I have 4.00 moles of a gas at a pressure of 5.60 atm and a volume of 12.0 liters, what is the temperature?

Example #2 - If I have 7.70 moles of gas at a pressure of 105 kPa and at a temperature of 56.0°C, what is the volume of the container that the gas is in?

Dalton's Law of Partial Pressures

In words:

P_{total} =

P_a =

X_a =

Example #1 - The total pressure of an O₂-Ar-He gas mixture is 755 mmHg. If the partial pressure of Ar is 174 mmHg and the partial pressure of He is 389 mmHg, then the partial pressure of O₂ is ---

Example #2 - The total pressure of an ideal gas is 2.0 atm. It is a mixture that is 20% oxygen and 80% nitrogen. What is the partial pressure of each gas in atm?

Example #3 - What's the partial pressure of carbon dioxide in a container that holds 5 moles of carbon dioxide, 3 moles of nitrogen, and 1 mole of hydrogen and has a total pressure of 1.05 atm?